

Exercises for Normalisation Forms

Abbreviations

1NF: First Normal Form

2NF: Second Normal Form

3NF: Third Normal Form

BCNF: Boyce Code Normal Form

FD: Functional Dependency

PK: Primary Key (ColumnName#)

CPK: Composite Primary Key (ColumnName#, ColumnName# ...)

FK : Foreign Key (ColumnName#::))

Exercise 1 - A EMPLOYEE spread sheet is given in EXCEL below to save their entries by their cards. We want to transfer this table in to a database. The primary key is Emp No. Apply NFs.

EmpNo	EmployeeName	TimeCardNo	TimeCardDate	DeptNo	DeptName
10	Thomas Arquette	106	11/02/2002	20	Marketing
10	Thomas Arquette	106	11/02/2002	20	Marketing
10	Thomas Arquette	106	11/02/2002	20	Marketing
10	Thomas Arquette	115	11/09/2002	20	Marketing
99	Janice Smitty			10	Accounting
500	Alan Cook	107	11/02/2002	50	Shipping
500	Alan Cook	107	11/02/2002	50	Shipping
700	Ernest Gold	108	11/02/2002	50	Shipping
700	Ernest Gold	116	11/09/2002	50	Shipping
700	Ernest Gold	116	11/09/2002	50	Shipping

Exercise 1 - 1NF:

- EMPLOYEE table should be arranged as follows according to the primary key in database systems
- The repeating group is Time Card No and Time Card Date by EmpNo

<u>EmpNo#</u>	EmployeeName	TimeCardNo	TimeCardDate	DeptNo#	DeptName
10	Thomas Arquette	106	11/02/2002	20	Marketing
		115	11/09/2002		
99	Janice Smitty			10	Accounting
500	Alan Cook	107	11/02/2002	50	Shipping
700	Ernest Gold	108	11/02/2002	50	Shipping
		116	11/09/2002		

Exercise 1 - 1NF :

EMPLOYEE			
<u>EmpNo#</u>	EmployeeName	DeptNo#	DeptName
10	Thomas Arquette	20	Marketing
99	Janice Smitty	10	Accounting
500	Alan Cook	50	Shipping
700	Ernest Gold	50	Shipping

- TimeCardNo is **unique** naturally. Therefore, there is no need to have a composite primary key on EmpNo and TimeCardNo.
- EmpNo is one foreign key

EMPLOYEE_CARD		
<u>EmpNo#</u>	<u>TimeCardNo#</u>	TimeCardDate
10	106	11/02/2002
10	115	11/09/2002
500	107	11/02/2002
700	108	11/02/2002
700	116	11/09/2002

Exercise 1 – 2NF : There is no composite key, so no 2NF.

Exercise 1- 3NF:

- DeptNo has a functional dependency (FD) with DeptName. So 3NF must be done on EMPLOYEE

EMPLOYEE		
<u>EmpNo#</u>	EmployeeName	DeptNo#
10	Thomas Arquette	20
99	Janice Smitty	10
500	Alan Cook	50
700	Ernest Gold	50

DEPARTMENT	
<u>DeptNo#</u>	DeptName
20	Marketing
10	Accounting
50	Shipping

- EMPLOYEE_CARD table does not have another FD except its PK. So no 3NF on this table.

Exercise 1- Final tables are :

EMPLOYEE		
<u>EmpNo#</u>	EmployeeName	DeptNo#
10	Thomas Arquette	20
99	Janice Smitty	10
500	Alan Cook	50
700	Ernest Gold	50

DEPARTMENT	
<u>DeptNo#</u>	DeptName
20	Marketing
10	Accounting
50	Shipping

N 1

1

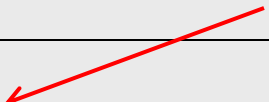
N

EMPLOYEE_CARD		
<u>EmpNo#</u>	<u>TimeCardNo#</u>	TimeCardDate
10	106	11/02/2002
10	115	11/09/2002
500	107	11/02/2002
700	108	11/02/2002
700	116	11/09/2002

Exercise 2- PART table is given below, apply NFs

PART	
<u>PART #</u>	WAREHOUSE
P0010	Warehouse A, Warehouse B, Warehouse C
P0020	Warehouse B, Warehouse D

Multi valued
Attribute



Better, but still flawed!



Really Bad Set-up!



PART 1			
<u>PART #</u>	WAREHOUSE A	WAREHOUSE B	WAREHOUSE C
P0010	Yes	No	Yes
P0020	No	Yes	Yes

Exercise 2 (continues)

- The table must be design as follows considering the step 6 of the mapping ER model to Relational Model.

PART2	
<u>PART #</u>	<u>WAREHOUSE#</u>
P0010	A
P0010	B
P0010	C
P0020	B
P0020	D

- CPK is on PART and WAREHOUSE
- PART2 is already in 1NF, 2NF and 3NF

Exercise 3 If PART table is given below, apply NFs

**Best
Form**



<u>PART</u>		
<u>PART #</u>	<u>WAREHOUSE #</u>	QUANTITY
P0010	Warehouse A	400
P0010	Warehouse B	543
P0010	Warehouse C	329
P0020	Warehouse B	200
P0020	Warehouse D	278

The table is already in 1NF, 2NF and 3NF

Exercise 4 - If the PART table is as follows, then it violates the 2NF

PART			
<u>PART #</u>	<u>WAREHOUSE #</u>	QUANTITY	WAREHOUSE ADDRESS
P0010	Warehouse A	400	1608 New Field Road
P0010	Warehouse B	543	4141 Greenway Drive
P0010	Warehouse C	329	171 Pine Lane
P0020	Warehouse B	200	4141 Greenway Drive
P0020	Warehouse D	278	800 Massey Street

Exercise 4 - After 2NF:

PART_WAREHOUSE		
<u>PART #</u>	<u>WAREHOUSE #</u>	QUANTITY
P0010	Warehouse A	400
P0010	Warehouse B	543
P0010	Warehouse C	329
P0020	Warehouse B	200
P0020	Warehouse D	278

N

1

WAREHOUSE	
<u>WAREHOUSE #</u>	WAREHOUSE_ADDRESS
Warehouse A	1608 New Field Road
Warehouse B	4141 Greenway Drive
Warehouse C	171 Pine Lane
Warehouse D	800 Massey Street

Exercise 5- EMP_DEPT table is given below. Apply NFs

EMP_DEPT				
<u>EMPNO#</u>	FIRSTNAME	LASTNAME	WORKDEPT#	DEPTNAME
000290	John	Parker	E11	Operations
000320	Ramlal	Mehta	E21	Software Support
000310	Maude	Setright	E11	Operations

- **EMP-DEPT** table doesnot have a repeating group, no 1NF
- It does not have a composite key, no 2NF
- **EMP-DEPT** table violates the 3NF below

Exercise 5 - 3NF:

EMP_DEPT			
<u>EMPNO#</u>	FIRSTNAME	LASTNAME	WORKDEPT#
000290	John	Parker	E11
000320	Ramlal	Mehta	E21
000310	Maude	Setright	E11

DEPT	
<u>WORKDEPT#</u>	DEPTNAME
E11	Operations
E21	Software Support

Diagram illustrating a 1:N relationship between EMP_DEPT and DEPT. A line connects the WORKDEPT# column in EMP_DEPT to the DEPT table, with a '1' near the EMP_DEPT table and an 'N' near the DEPT table.

Exercise 6 - STUDENT table is given below, apply NFs

STUDENT						
<u>Student#</u>	Student FName	Student LName	Advisor#	Advisor	Adv-Room	Class#
1022	Jane	Mayo	10	Susan Jones	412	101-07 143-01 159-02
4123	Mark	Baker	12	Anne Smith	216	101-07 159-02 214-01

Really Bad Set-up!

**Multi valued
Attribute**

14 of 30

Exercise 6 (continues)- STUDENT Table is rearranged according to the Multi-valued attribute of Class.

- Considering the step 6 of the mapping ER model to Relational Model, the table must be designed as follows:

STUDENT1				
<u>Student#</u>	Advisor#	AdvFirst Name	AdvLast Name	Adv-Room
1022	10	Susan	Jones	412
4123	12	Anne	Smith	216

STUDENT_CLASS	
<u>Student#</u>	<u>Class#</u>
1022	101-07
1022	143-01
1022	159-02
4123	201-01
4123	211-02
4123	214-01

Now the both table is already in 1NF

Exercise 6 (continues) - 2NF -

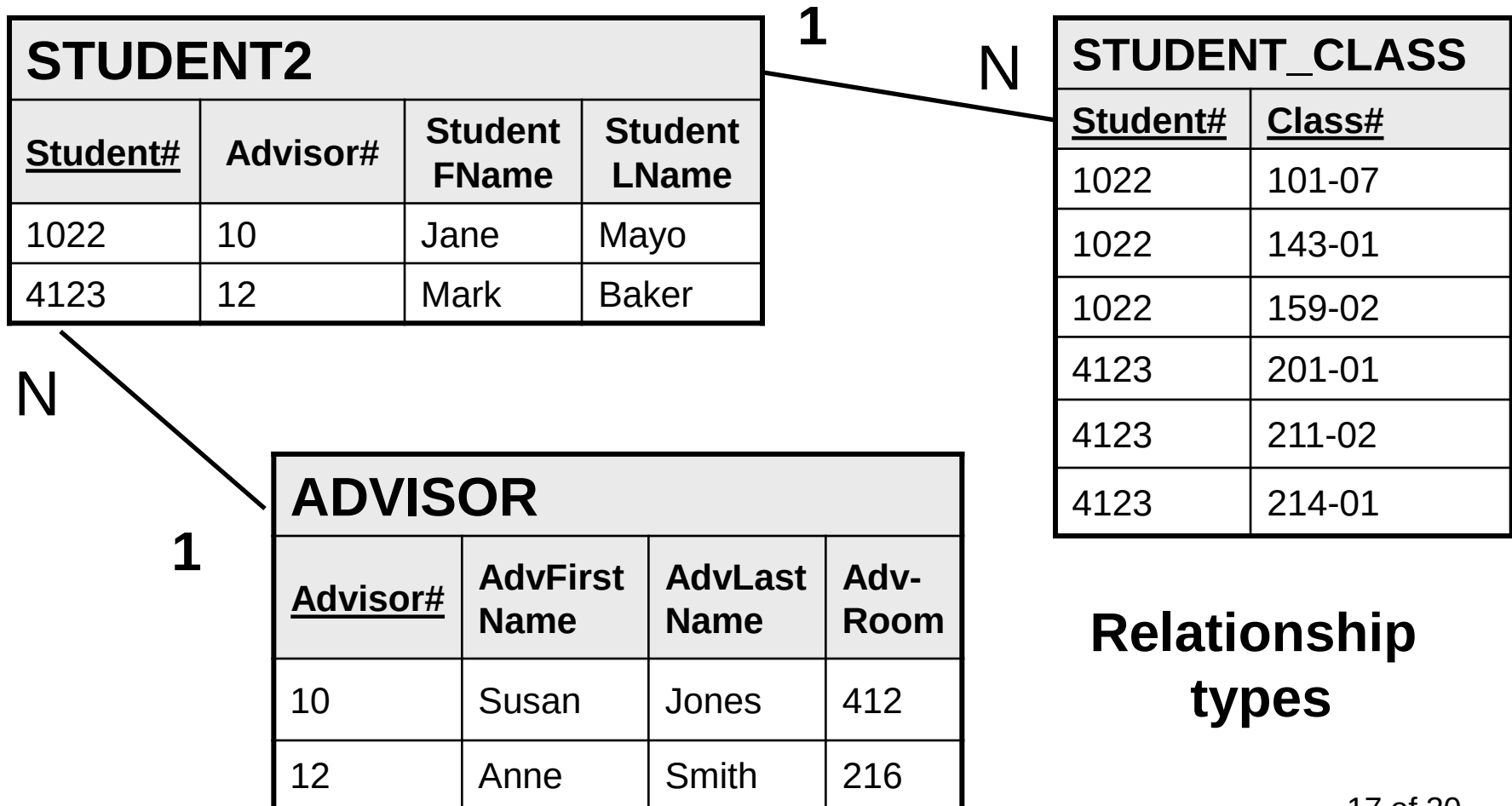
- STUDENT 1 table doesnot have a CPK, therefore no 2NF.
- STEDENT_CLASS table has a CPK but it doesnot have another column which has a FD.

Therefore the both tables are in 2NF

Exercise 6 - 3NF of STUDENT1 Table

Another FD on STUDENT1 Table is

Advisor $\rightarrow$ AdvFirstName, AdvLastName, Adv-Room



Exercise 7 - Un-normalized table is given below, apply NFs.

One employee works on different projects for the certain time percentages.

EMPLOYEE									
<u>EmpID#</u>	Name	Dept Code#	Dept Name	Proj 1	Time Proj 1	Proj 2	Time Proj 2	Proj 3	Time Proj 3
EN1-26	Sean Breen	TW	Technical Writing	30-T3	25%	30-TC	40%	31-T3	30%
EN1-33	Amy Guya	TW	Technical Writing	30-T3	50%	30-TC	35%	31-T3	60%
EN1-36	Liz Roslyn	AC	Accounting	35-TC	90%				

1st project 2nd project 3rd project

The repeating group consists of project number and time on an employee and therefore the table has to be rearranged.

Exercise 7 (continues)-

Arranged table:

EMPLOYEE1						
<u>EmpID#</u>	Last Name	First Name	Dept Code#	Dept Name	Project Number	Time on Project
EN1-26	Breen	Sean	TW	Technical Writing	30-T3	25%
					30-TC	40%
					31-T3	30%
EN1-33	Guya	Amy	TW	Technical Writing	30-T3	50%
					30-TC	35%
					31-T3	60%
EN1-36	Roslyn	Liz	AC	Accounting	35-TC	90%

Exercise 7 (continues)-

Now we will do the 1NF:

EMP_PROJ		
<u>EmpID#</u>	<u>Project Number#</u>	Time on Project
EN1-26	30-T3	25%
EN1-26	30-T3	40%
EN1-26	31-T3	30%
EN1-33	30-T3	50%
EN1-33	30-TC	35%
EN1-33	31-T3	60%
EN1-36	35-TC	90%

EMPLOYEE2				
<u>EmpID#</u>	Last Name	First Name	Dept Code	Dept Name
EN1-26	Breen	Sean	TW	Technical Writing
EN1-33	Guya	Amy	TW	Technical Writing
EN1-36	Roslyn	Liz	AC	Accounting

- EMP_PROJ has one functional dependency on EmpID# and Project Number#, therefore there is no 2NF.
- EMPLOYEE2 does not have a composite key, therefore there is no 2NF.

Exercise 7 (continues)-

Tables in 3NF

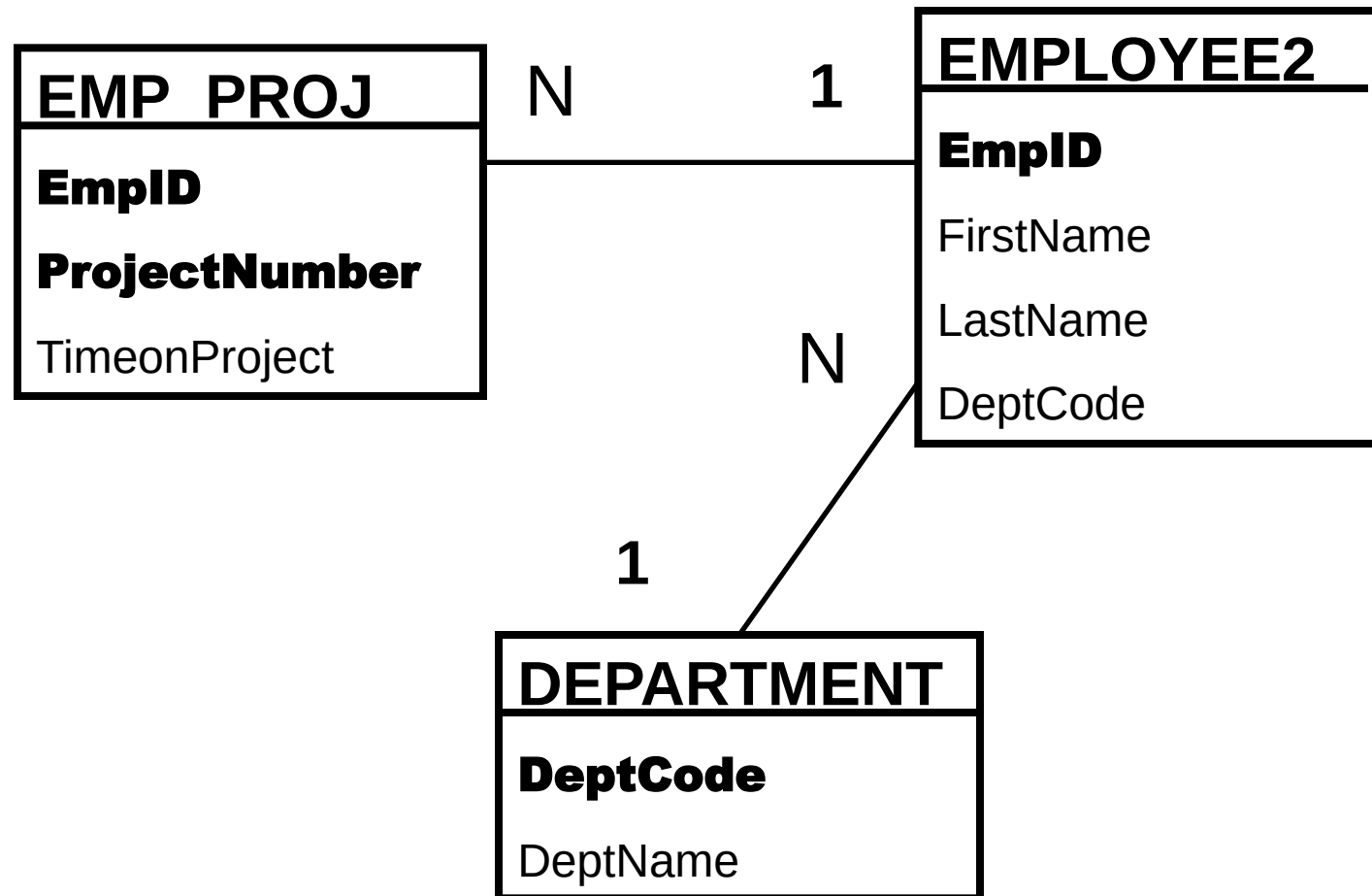
EMP_PROJ		
<u>EmpID#</u>	<u>Project Number#</u>	Time on Project
EN1-26	30-T3	25%
EN1-26	30-T3	40%
EN1-26	31-T3	30%
EN1-33	30-T3	50%
EN1-33	30-TC	35%
EN1-33	31-T3	60%
EN1-36	35-TC	90%

EMPLOYEE2			
<u>EmpID#</u>	Last Name	First Name	Dept Code#
EN1-26	Breen	Sean	TW
EN1-33	Guya	Amy	TW
EN1-36	Roslyn	Liz	AC

DEPARTMENT	
<u>DeptCode#</u>	Dept Name
TW	Technical Writing
AC	Accounting

Exercise 7 (continues)-

Relationship types:



Exercise 8: Un-normalized table, EMPLOYEE is given below, apply NFs

EMPLOYEE								
<u>Emp ID#</u>	FName	LName	Dept Manager ID#	Dept Manager Name	Dept No#	DeptName	Sector	Spouse Or Children
285	Carl	Carlson	1	Smithers	EN	Engineering	6G	
365	Lenny	Clark	1	Smithers	MK	Marketing	8G	
458	Homer	Simpson	2	Mr. Burns	SF	Safety	7G	Marge, Bart, Lisa, Maggie

Multi-Valued Attribute

Really Bad Set-up!

23 of 30

Exercise 8 (continues) :

EMPLOYEE table can be design as follows:

EMPLOYEE1											
<u>Emp ID#</u>	FName	LName	Dept Manager ID#	Dept Manager Name	Dept Code#	Dept Name	Dept Sector	Spouse	Child 1	Child 2	Child 3
285	Carl	Carlson	1	Smithers	EN	Engineering	6G				
365	Lenny	Clark	1	Smithers	MK	Marketing	8G				
458	Homer	Simpson	2	Mr. Burns	SF	Safety	7G	Marge	Bart	Lisa	Maggie

Better, but still flawed!

1st dependent

2nd dependent

3rd dependent

4th dependent

Exercise 8 (continues) :

EMPLOYEE table can also be design as follows:

EMPLOYEE2								
<u>Emp ID#</u>	FName	LName	Dept Manager ID#	Dept Manager Name	Dept Code#	Dept Name	Dept Sector	Dependent
285	Carl	Carlson	1	Smithers	EN	Engineering	6G	
365	Lenny	Clark	1	Smithers	MK	Marketing	8G	
458	Homer	Simpson	2	Mr. Burns	SF	Safety	7G	Marge
458	Homer	Simpson	2	Mr. Burns	SF	Safety	7G	Bart
458	Homer	Simpson	2	Mr. Burns	SF	Safety	7G	Lisa
458	Homer	Simpson	2	Mr. Burns	SF	Safety	7G	Maggie

Better, but still flawed!

- Spouse/Children attribute's type is the multi-valued attribute and there is no repeating group (a single column value with the same data type).
- The table must be designed as two tables by the step 6 of the mapping ER model to Relational Model,
- The tables must be designed as follows:

Exercise 8 (continues) :

EMPLOYEE3							
<u>EmpID#</u>	FName	LName	Dept ManagerID#	Dept Manager Name	Dept Code#	Dept Name	Dept Sector
285	Carl	Carlson	1	Smithers	EN	Engineering	6G
365	Lenny		1	Smithers	MK	Marketing	8G
458	Homer	Simpson	2	Mr. Burns	SF	Safety	7G

EMP_DEP	
<u>EmpID#</u>	<u>Dependent#</u>
458	Marge
458	Bart
458	Lisa
458	Maggie

← **Good Set-up!**

Both tables now are in 1NF and 2NF.

Exercise 8 (continues) – 3NF

- EMP-DEP does not have the second FD, stays the same
- Another FD on EMPLOYEE3 table is

DeptCode $\rightarrow$ DeptName, DeptSector, DeptManagerID#,
DeptManagerName

EMPLOYEE4			
<u>EmpID#</u>	FName	LName	DeptCode#
285	Carl	Carlson	EN
365	Lenny	Clark	MK
458	Homer	Simpson	SF

DEPARTMENT1				
<u>DeptCode#</u>	DeptName	DeptSector	DeptManagerID#	DeptManagerName
EN	Engineering	6G	2	Smithers
MK	Marketing	8G	2	Smithers
SF	Safety	7G	1	Mr. Burns

Exercise 8 (continues) – 3NF

- But still there is a FD on DEPARTMENT 1 as follows:
DeptManagerID# → DeptManagerName
and we need to apply the 3NF one more

EMPLOYEE4			
<u>EmpID#</u>	FName	LName	DeptCode#
285	Carl	Carlson	EN
365	Lenny	Clark	MK
458	Homer	Simpson	SF

MANAGER	
<u>ManagerID#</u>	ManagerName
1	Mr. Burns
2	Smithers

DEPARTMENT2			
<u>DeptCode#</u>	DeptName	DeptSector	DeptManagerID#
EN	Engineering	6G	2
MK	Marketing	8G	2
SF	Safety	7G	1

Exercise 8— Final tables with relational types after NFs are :

EMPLOYEE4			
<u>EmpID#</u>	FName	LName	DeptCode#
285	Carl	Carlson	EN
365	Lenny	Clark	MK
458	Homer	Simpson	SF

1 N

EMP_DEP	
<u>EmpID#</u>	<u>Dependent#</u>
458	Marge
458	Bart
458	Lisa
458	Maggie

N

1

DEPARTMENT2			
<u>Dept Code#</u>	Dept Name	Dept Sector	Dept Manager ID#
EN	Engineering	6G	2
MK	Marketing	8G	2
SF	Safety	7G	1

N

1

MANAGER	
<u>Manager ID#</u>	Manager Name
1	Mr. Burns
2	Smithers

Exercise 9: Un-normalized table, VISIT is given below, apply NFs

VISIT										
<u>Rep ID#</u>	Representer	Client ID1	Client 1	Time 1	Client ID2	Client 2	Time 2	Client ID3	Client 3	Time 3
TS-89	Gilroy Gladstone	978	US Corp.	14 hrs	665	Taggarts	26 hrs	782	Kilroy Inc.	9 hrs
RK-56	Mary Mayhem	221	Italiana	67 hrs	982	Linkers	2 hrs			

- Note that the representer can be designed with RepFirstName and RepLastName as two columns for a better design if you wish or can remain the same.
- However the table has to be rearranged as follows for repeating group of clients based on the PK of RepID.

Exercise 9 (continues):

<u>RepID#</u>	RepFirstName	RepLastName	ClientID#	Client	Time
TS-89	Gilroy	Gladstone	978	US Corp	14 hrs
			665	Taggarts	26 hrs
			782	Kilroy Inc.	9 hrs
RK-56	Mary	Mayhem	221	Italiana	67 hrs
			982	Linkers	2 hrs

Repeating group (more than one column with different data types) of ClientID, Client and Time for each key

Complete NFs as an exercise 😊)))